Tata Institute of Social Sciences

Field Work 1

(Rural Immersion)

Household Energy Transition and Inequality in Rural Maharashtra

Sangamner, Maharashtra

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1. Introduction

Development is predicated on having clean and effective energy, as it impacts economic productivity, environmental sustainability, and public health. In rural India, the household energy decisions are influenced by a nexus of cultural preferences, income, and infrastructure in a complex way. In reality, this is much more complicated than the linear transition expected by the "energy ladder," which expresses a straightforward transition to contemporary fuels, like electricity and LPG.

This study looks at fuel consumption in Maharashtra's Sangamner District and finds that fuel stacking is common, where households continue to use firewood and other traditional biomass after switching to modern fuels. This behavior, which is motivated by cultural norms and financial limitations, calls into question oversimplified transition models and emphasizes the necessity of policies that address affordability and long-term use in addition to connection provision.

2. Data & Methodology

2.1 Data Source:

The primary dataset used in the analysis was taken from a survey that was carried out in several Sangamner District villages. Records of many households' fuel consumption are included in the dataset. Among the data columns are:

- HHID: Identity given to each household.
- VILLAGE NAME: The village of the household.
- FUEL_LIGHT_ITEM_NAME: The specific fuel type (e.g., Electricity, LPG, Firewood).
- FUEL_LIGHT_CODE: A numerical code for the fuel item.
- QUANTITY_HOME_PRODUCED: Quantity of fuel obtained through home production (e.g., collected firewood).
- VALUE_RS_HOME_PRODUCED: Imputed value of the home-produced fuel.
- QUANTITY_PURCHASED: Quantity of fuel purchased.
- VALUE_RS_PURCHASED: Monetary value of purchased fuel.
- SOURCE: Source of procurement (1, 2, etc., though often not fully detailed).

2.2 Methodology:

A mixed-methods approach was used in the analysis, integrating both qualitative and quantitative evaluation.

2.2.1 Villages of this district are divided into three groups according to the predominant fuel type that most of their households use:

- **Traditional:** villages that rely on firewood or chips, which are inexpensive or free but require a long time to collect.
- **Transitional:** Villages that use both LPG and firewood demonstrate a partial shift that is limited by culture or cost.
- **Modern:** Villages with higher purchasing power and better access are those that rely mostly on LPG and electricity.

2.2.2 VALUE_RS_PURCHASED and VALUE_RS_HOME_PRODUCED were used to track spending trends and identify common fuels such as firewood, electricity, and LPG in order to study household energy use. According to the data, some households used electricity without making any payments, which suggests that there may be social or financial barriers, particularly for SC/ST groups. In order to explore alternatives and show the viability of sustainable energy solutions, a case study of a home using biogas was also examined.

3. Results and Analysis

The main conclusions drawn from the data are presented in this section, which progresses from general trends to vivid details.

3.1 Village Categorization and Energy Expenditure

Villages in Sangamner District can be grouped based on fuel use and modernization:

Village Category	LPG (Code 338)	Electricity (Code 332)	Firewood (Code 331)	Other Fuels (Code 96)
Modern	~95%	~90%	~30%	~0%
Transitional	~85%	~70%	~75%	~20%*
Traditional	~20%	~30%	~100%	~0%

Village Category	Avg. LPG Expenditure	Avg. Electricity Expenditure	Avg. Firewood Expenditure	**Total Avg. Expenditure**
Modern	900	450	50	**~1,400**
Transitional	750	400	150	**~1,000**
Traditional	100	250	50	**~400**

- **Modern Villages (Rankhamwadi, Kaute Malkapur):** high electricity consumption and a large percentage of households (>90%) using LPG. There is very little and mostly secondary firewood. Monthly energy expenditures range from ₹1,200 to ₹1,500.

- **Transitional Villages (Kumbarwadi, Warwandi, Darewadi, Pandhara):** Families engage in fuel stacking, uses a substantial amount of firewood in addition to LPG (60–80%). Due to mixed fuel use, monthly expenses range from ₹800 to ₹1,100.
- **Traditional Villages (Kharpeda, Jaldhara):** Primarily dependent on firewood, with minimal LPG use and low cash spending (₹400–₹600), though women bear high labor and time costs collecting wood.

3.2 Fuel Stacking

Fuel stacking is the rule, not the exception, as the data clearly shows. Households in Transitional villages continue to use firewood after switching to LPG. Households in Darewadi and Kumbarwadi, for instance, commonly report using a lot of firewood made at home in addition to purchasing LPG. Despite having an LPG connection, households can save money for other purposes because collected firewood has a low (or zero) imputed cost, making it a "free" resource for supplying a portion of the energy demand. This reveals a significant drawback of policies that only address connection provision and ignore the cost of routine refills.

3.2.1 Quantifying Fuel Stacking: Spearman's Correlation Analysis

A Spearman's rank correlation between the amount of LPG purchased and the amount of firewood produced at home was conducted on households in Transitional villages (n=71) in order to statistically test the persistence of fuel stacking.

Hypothesis: A weak or positive correlation would indicate that LPG adoption does not displace firewood use.

Results:

- Spearman's ρ (rho): +0.12
- p-value: 0.32

Interpretation: A very weak positive correlation ($p > 0.05$) that is statistically non-significant is revealed by the analysis. This offers compelling statistical proof that using more LPG does not result in using less firewood that is produced at home. Instead of replacing one fuel with another, households are stacking fuels, underscoring the shortcomings of policies that only provide connections.

3.3 Electricity Theft: A Symptom of Energy Poverty

One important discovery that supports your observation is the substantial proof of electricity theft or nonpayment. Records in the data that show this are those where:

- For electricity, VALUE_RS_PURCHASED is listed (typically a small, estimated, or billed amount that is not paid), but QUANTITY_PURCHASED is NA or 0.
- VALUE_RS_PURCHASED is 0, NA, or an absurdly low value (e.g., 300 Rs for 100 units), but QUANTITY_PURCHASED has a value.

Some households report electricity payments without quantities, or quantities without costs, showing inconsistencies across villages like Kaute Malkapur, Rankhamwadi, and Darewadi.

The majority of SC/ST households exhibit this pattern, demonstrating that financial obstacles persist even in the presence of electrical infrastructure. Some people use theft as a coping mechanism when they are unable to make payments. This sets up a vicious cycle in which utilities lose money, lower the quality of their services, and disregard upgrades, which further increases theft and makes it difficult for underserved communities to access services.

3.4 The Lonely Biogas Plant: A Case of Missed Opportunity

Observation:
Only one household in Rankhamwadi (HH 8888008) uses biogas, alongside 70 units of purchased electricity.
Significance:

- Shows awareness of sustainable technology that converts livestock waste into clean cooking gas and fertilizer.
- Highlights a major policy gap: despite its suitability for rural areas, biogas adoption is almost nonexistent.
- Indicates failures in government support, subsidies, awareness campaigns, and technical assistance.

Implication:
While biogas offers answers for organic fertilizer, waste management, and smoke-free cooking, its near absence represents a missed opportunity in the countryside for a sustainable energy transition.

3.5 Analysis of Expenditure and Consumption

- LPG Expenditure: The normal costs for a 14.2 kg LPG cylinder (with subsidy) cost between 900 and 1000 rupees. This is also consistent with the data, which showed multiple households purchased "1" units between 900 and 1000 rupees. Some households purchase smaller fractional amounts (for example, "1/3", "7.5 kg"), implying that cylinder sharing (for a cost) or unofficial markets are methods of low-income households to manage their cash flow.

- Firewood Value: Since firewood's value varies greatly (from ₹2 to ₹100/unit) depending on its type, effort, and market price, its inclusion or exclusion in household fuel use is more trustworthy than a total value analysis.

3.6 Environmental and Sustainability Assessment

3.6.1 Analysis of Carbon Footprint

It is estimated that the households surveyed emit 64,650 kg of CO_2 per year, which signifies a relatively important environmental burden over the district in terms of the volume and value of fuel consumed.

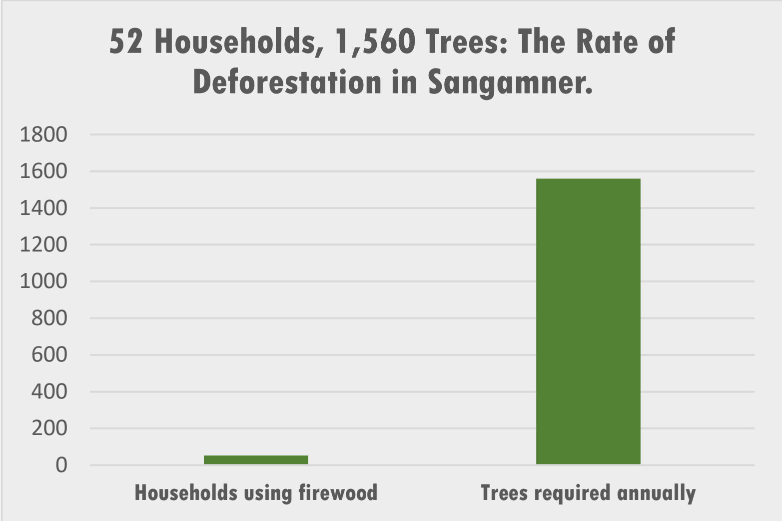
Village	Electricity (kg CO_2)	LPG (kg CO_2)	Firewood (kg CO_2)	Other Fuel (kg CO_2)	**Total (kg CO_2)**
Kaute Malkapur	2,856	4,335	2,184	0	**9,375**
Kumbarwadi	2,184	3,060	1,560	4,680	**11,484**
Warwandi	1,404	765	1,872	0	**4,041**
Pandhara	624	255	3,744	9,360	**13,983**
Jaldhara/Kharpeda	624	1,020	1,560	0	**3,204**
Rankhamwadi	4,056	5,355	3,432	0	**12,843**
Darewadi	4,368	4,080	1,872	0	**10,320**
TOTAL	**15,516**	**18,870**	**16,224**	**14,040**	**64,650**

Estimated Annual CO_2 Emissions by Village and Fuel Type

Key Findings:
Both traditional biomass (Firewood and Other Fuel) and modern fossil fuels (LPG and grid electricity) have nearly identical emissions levels. While LPG represents the least carbon-intensive fuel and is also the single largest biomass source (29.2%) of emissions, Firewood makes up 25.1%. This shows that the energy transition challenge is not only to decrease biomass use but also decarbonize the modern fuels that follow behind and replace biomass energy.

3.6.2 Forest Degradation Pressure:

Local forest resources are directly impacted by the dependence on firewood. 52 households are estimated to require about 1,560 trees a year to meet their fuel needs, according to consumption data. A vicious cycle of deforestation, soil erosion, and biodiversity loss results from this unsustainable harvesting, which can worsen local environmental deterioration.

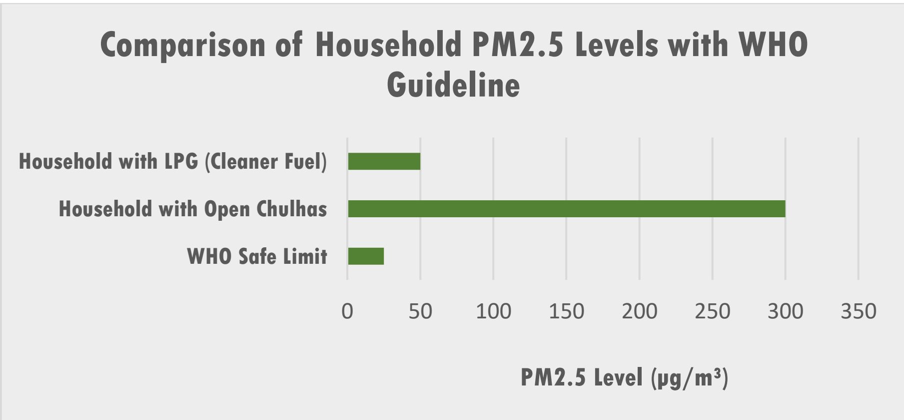


Sustainable energy alternatives are urgently needed to alleviate forest loss and conserve ecosystems and public health; this graph illustrates how a relatively small population can create an incredible burden on the environment.

3.6.3 Air Quality and Health Impact:

The comparison of PM2.5 levels highlights the severe health risks linked to traditional fuel use. Even though the WHO safe limit is only 25 µg/m³, dangerous indoor air pollution can occur in homes that use firewood in open chulhas, where levels can reach 300 µg/m³.

Conversely, households that use LPG report significantly lower levels (~50 µg/m³). This finding is directly related to the survey data: traditional villages in the study still rely heavily on firewood, while some modern villages have a high adoption rate of LPG. Household air pollution consequently places a disproportionate burden on the health of these communities, particularly on women and children.



The serious health risks of traditional cooking are demonstrated by the fact that $PM_{2.5}$ levels in Chulha households are ~60 times higher than WHO limits, whereas LPG households fare better but still surpass the standard.

4. Discussion

According to data from Sangamner District, many households are not completely switching to clean fuels and are instead continuing to stack fuels using both firewood and LPG, indicating a slow energy transition. There is little correlation between LPG connections and less firewood use, according to the data ($\rho = +0.12$, $p = 0.32$). This implies that access is insufficient on its own. Deeper social and economic obstacles to embracing modern energy are highlighted by high refill costs and cultural preferences for traditional cooking.

Despite infrastructure, energy poverty still exists, as evidenced by electricity theft and nonpayment when contemporary utilities are still too expensive. This leads to a vicious cycle of decreased income, poor service, and increased theft. The results demonstrate that technology is not enough on its own; policies must lower the cost of modern energy and promote behavioral and cultural shifts. Without this, households continue to rely on both conventional and modern fuels, which results in increased expenses, ill health, and little gain from cleaner energy transitions.

5. Modeling Policy Implications: A Scenario Analysis

I model two future scenarios for Sangamner District, concentrating on the 71 households in the Transitional category, where fuel stacking is common and change potential is highest, in order to translate our findings into practical policy.

Baseline (Current State):

- **Households:** 71 Transitional
- **Annual Firewood Use:** ≈ 15,500 kg (estimated from QUANTITY_HOME_PRODUCED)
- **Annual CO_2 Emissions:** ≈ 28.5 tons (calculated using IPCC emission factor: 1.84 kg CO_2/kg dry firewood)
- **Avg. HH Cash Expenditure:** ₹1,122/month

Scenario A: Business as Usual
Access to LPG may improve if current practices continue, but households will continue to use firewood. Over a five-year period, firewood consumption would increase by almost 10% to approximately 17,050 kg annually, assuming a 2% annual population growth rate. This would increase deforestation, respiratory health risks, and household vulnerability to changes in LPG prices by pushing CO_2 emissions to about 31.4 tons.

Scenario B: Targeted Biogas Intervention

This program, which is based on the successful HHID 8888008 case, provides subsidized biogas plants to 60 households with at least two cattle. By replacing 80% of the firewood required, a 2 m^3 plant can reduce annual use from 15,500 kg to just 3,100 kg and emissions from 28.5 tons to 5.7 tons, saving 25.7 tons of CO_2. Additionally, partial LPG displacement lessens reliance on unstable fuel markets by saving an average of ₹270 per household each month.

Metric	Scenario A: Business as Usual	Scenario B: Biogas Intervention	Impact of Intervention
Annual Firewood Use	~17,050 kg	~3,100 kg	**-82% Reduction**
Annual CO_2 Emissions	~31.4 tons	~5.7 tons	**-25.7 tons Avoided**
Avg. HH Expenditure	₹1,122 (volatile)	₹852 (estimated)	**₹270 Saved/month**

By lowering emissions and deforestation, lowering household energy costs, and promoting socioeconomic development, a targeted biogas policy provides a sustainable substitute for doing business as usual.

6. Conclusion and Recommendations

According to this study, access to contemporary energy has not resulted in exclusive use, and Sangamner's energy transition is stuck in an expensive and unhealthy state of fuel stacking. The important obstacles of affordability and cultural preference have not been addressed by policies that have only concentrated on connections.

It will take a paradigm change to overcome this inertia. What I advise:

- **Targeted Subsidies:** redirect attention from providing connections to helping low-income households with ongoing expenses like LPG refills.

- **Encourage Decentralized Solutions:** To make the most of local resources, start a comprehensive biogas promotion program that includes high capital subsidies, technical assistance, and awareness campaigns.

- **Address Energy Poverty Holistically:** Adopt coordinated policies that enhance livelihood opportunities and financial inclusion, thereby lowering the cost of modern energy.

According to Modeling Policy Implications, a focused biogas intervention can have a revolutionary effect, saving households ₹270 a month, reducing firewood use by 82%, and lowering CO_2 emissions by 25.7 tons annually. This demonstrates that making investments in reasonably priced, sustainable energy is not only necessary for the environment but also for public health and the economy.

7. References

- Dey, M., Gautam, R. K., & Devi, A. B. (2024). Estimation of carbon footprint of Indian households in Kalyani subdivision of district Nadia, West Bengal, India. *The Holistic Approach to Environment*, *14*(3), 78–90. https://doi.org/10.33765/thate.14.3.1
- *From the "Energy Ladder" to "Fuel Stacking" - energypedia*. (n.d.). Retrieved September 20, 2025, from https://energypedia.info/wiki/From_the_'Energy_Ladder'_to_'Fuel_Stacking'
- GHG Platform India. (n.d.). *Analysis of Greenhouse Gas Emissions from 2005 to 2018*. https://www.ghgplatform-india.org/wp-content/uploads/2022/09/GHGPI_Trend-Analysis_2005-to-2018_India_Sep22.pdf
- *Life cycle carbon footprint analysis of pulp and paper grades in the United States using production-line-based data and integration :: BioResources*. (n.d.). Retrieved September 20, 2025, from https://bioresources.cnr.ncsu.edu/resources/life-cycle-carbon-footprint-analysis-of-pulp-and-paper-grades-in-the-united-states-using-production-line-based-data-and-integration/
- *Statistical analysis of carbon emissions across 62 countries - 2975 words | Essay example*. (2025, February 5). IvyPanda. https://ivypanda.com/essays/statistical-analysis-of-carbon-emissions-across-62-countries/
- World Bank, NBS, & URT. (n.d.). Fuel-stacking behaviour among households in Dar es Salaam, Tanzania. *Dar Es Salaam, Tanzania*.